WA VERIFY USABILITY REPORT

DOH CONTRACT HED25742, 03 AMENDMENT

JUNE 9, 2023

Executive Summary 3
Introduction 5
Objectives 5
Methods 5
Usability Test Environment 6
Recruitment 6
Sample Characteristics 6
Usability Session Instrument and Tasks 8
Data Collection and Analysis 9
Results 9
Participant Characteristics 9
    Vaccine Verification Method 9
    Access to Another Person's Vaccine Record 11
WA Verify Usability Exercise Results 11
    Task Completion 11
    QR Code Storage Method 12
    Future Use of WA Verify 13
Recommendations 13
    1. Clarify the format of WA Verify (App vs. Web Page) 14
    2. Improve readability in introductory paragraphs on the WA Verify website 15
    3. Provide information about where the immunization data comes from 15
    4. Provide information about average wait times to receive message with link (email and text) 16
    5. Provide more information/options for saving the QR code 17
    6. Clearly identify incoming text messages as coming from DOH/WA Verify 18
    7. Address problems displaying large numbers of vaccines 19
Conclusion 20
APPENDIX 21
Appendix A. WA Verify UX Questionnaire Recruitment Email 21
Appendix B. Screening Questionnaire 22
Appendix C. Test Script 24
Appendix D. Participants 29
Appendix E. Session Notes 30
Appendix F. Page Definitions 31

Executive Summary

The aim of this exercise was to assess the usability of the current WA Verify system and generate suggestions for improvement to this and future iterations of the system. An online (Zoom) usability exercise was developed and pilot tested by two individuals prior to recruitment. Usability exercise participants were recruited from a group of 2021 WA Notify survey respondents who indicated they were interested in participating in future evaluations of digital public health tools. Of the 1724 survey respondents contacted, 765 were eligible for the usability assessment, 39 were purposively selected for participation and 10 scheduled and completed the usability session.

Including the pilot testers, a total of twelve online usability sessions were completed, of which 6 participants were "new" users and 6 were current WA Verify users. Eleven participants completed all steps to sign up for WA Verify with minimal moderator assistance. One participant needed significant assistance due to a record matching error, not the user interface. All new users indicated they planned to adopt WA Verify as their primary vaccination verification method.

The usability issues identified through the exercise were minor and consisted of text content and formatting issues on the WA Verify website and text message communications. The primary recommendations to improve the usability of the WA Verify tool include:

- Overall communications: Clarify information and FAQs regarding the WA Verify platform, how to enroll and how it works, i.e., WA Verify is not an app but rather a website for enrollment and the QR code is stored on a phone.
- Website:
    - Improve readability by reducing the text density or reformatting the content as bullet points or visual displays. Include the phrase “Washington State Immunization Information System” in the website’s introductory paragraph and provide more information on how immunization information is saved, accessed, and stored on the FAQ page.
    - Improve user expectations by adding information regarding the average time to receive a text or email in both the introduction and confirmation sections of the website.
    - Inform users that a screenshot is an additional method of saving the QR code.
- Ensure authority of the SMS message by highlighting the source, i.e., the message is sent from WA Verify and the Washington State Department of Health.
- Reformat the downloadable version of the WA Verify QR code to include room for multiple boosters and continue testing to determine the extent to which formatting issues with multiple boosters affect users.

Findings from this usability exercise suggest that the WA Verify tool is easy to use, functional, and could be improved with a few minor changes and increased outreach. Most of the recommendations from this exercise will be applicable to expanded public health uses cases building on SMART Health Cards or SMART Health Links technology. As exploration of additional use cases continues, exercises like this will be helpful in gathering contextual information and understanding the perspective of users. Overall,

these findings will help inform the development and design of more usable public health digital tools.

Introduction

The Washington State Department of Health adopted WA Verify, a Verifiable Clinical Information (VCI) system, in October 2021. WA Verify allows users to quickly find, download and display their up-to-date COVID-19 Vaccination status in human-readable and Quick Response (QR) code formats. The tool was rapidly developed and implemented using new and existing components and includes a mobile-friendly interface. Usability was not a primary focus during WA Verify's rapid development and implementation; thus, this Usability Exercise aims to answer questions about the usability of the current WA Verify tool and inform recommendations aiming to improve this and future iterations of the system.

This Usability Exercise is an activity of Objective 5 (Usability) of the Evaluation & Technical Assistance Project: WA Notify-2-WA Verify (DOH CONTRACT HED25742, 03 AMENDMENT). The project is being conducted by an evaluation team at the University of Washington School of Public Health on behalf of the WA State Department of Health.

Objectives

The primary objectives of this small-scale usability exercise were to gather data from users or potential users of WA Verify in order to:

- Assess ease of use, satisfaction, acceptability, challenges to utilization, and benefits and drawbacks of using WA Verify;
- Identify commonly encountered obstacles to completing tasks with WA Verify; and
- Identify and categorize low, medium, and high priority usability issues and generate recommendations for addressing these issues.

To meet these objectives, the following questions guided the usability instrument development, task and session design:

- How do those new to and those who have adopted WA Verify experience completing standard tasks with the tool?
- After completing the setup process, how satisfied are users, generally, with WA Verify?
- In what ways do users find the tool helpful, useful, relevant, or frustrating?
- What challenges do users experience that prevent them from completing common WA Verify tasks?
- What improvements could be made to ensure users can complete common tasks more easily and faster? What improvements might be made to ensure the overall user experience is more satisfying?

Methods

Two instruments were developed to gather data from a small sample of new and experienced WA Verify users: A brief online screening questionnaire and an online (attached to this report as Appendix B), moderated task-focused usability session script (attached as Appendix C).

Usability Test Environment

The usability exercise sessions took place online using Zoom. The set-up required participants to use two internet-connected devices, one of which was a mobile phone, during the session. With participant verbal consent, sessions were recorded in Zoom solely to facilitate notetaking.

Recruitment

Participants were recruited from a sample of individuals who indicated interest in participating in future public health digital tool evaluations on a 2021 WA Notify Travel Survey which focused on travel experiences (N=1724). Invitations to complete the screening questionnaire were distributed to this group via email. The email described the usability activity as well as the $50 Amazon Gift Card incentive that selected participants would receive after completing the activity. The complete recruitment email is included as an attachment (Appendix A) to this report. Eligibility criteria for participation in the usability exercise included: current WA State resident, receiving at least one dose of a COVID-19 vaccine in WA state, owning a smartphone, and had a mobile phone and one other device available for use during the usability session.

Sample Characteristics

A total of 808 individuals completed the screening questionnaire and 765 met the eligibility requirements. A group of 39 eligible respondents were purposively selected for recruitment to represent the following demographic categories:

- Geographic Location: Eastern vs. Western WA
- User Status: new vs. existing user
- Age: under 30, 30-59, or over 60 years
- Smartphone OS: Apple vs. Android

Recruitment was conducted by email with invitations to schedule a 30-minute usability session with the moderator distributed over eight successive waves.

Table 1, Figure 1 and *Figure 2* show the distribution of both individual traits and the distributions between the characteristics of interest of the final usability session participants were near the original goals. A listing of all participants and their Screening Questionnaire responses is included in Appendix D.

Table 1. Final Usability Session Participant Goals and Actual Characteristics

	Geography		**User Status**		**Age**			**Smartphone OS**		**Gender**			
	East	West	New	Existing	>30	30 - 59	60+	Apple	Android	M	F	NB	Total
Goal	**6**	**6**	**6**	**6**	**4**	**4**	**4**	-	-	-	-	-	12
Participants	6	6	6	6	4	5	3	6	6	1	9	2	12

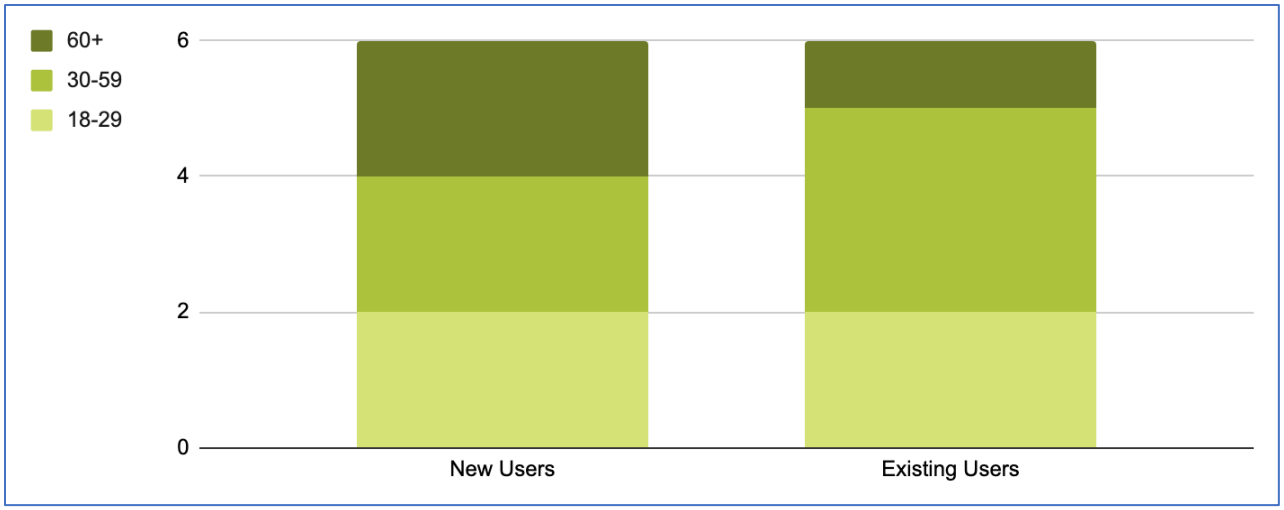


Figure 1. Age Groups of New and Existing Users, Final Participants

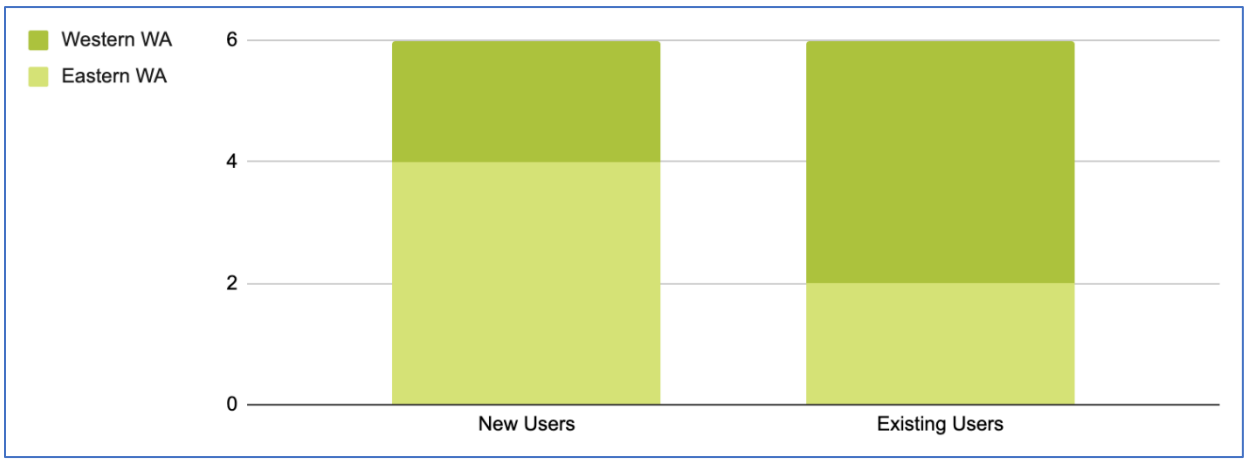


Figure 2. Location Distribution of New and Existing Users, Final Participants

Purposive selection focused on balancing Geographic location (East/West), User Status (New/Existing) and Age (under 30, 30-59, 60+) characteristics, led to an uneven distribution of other demographic variables. For example, gender is skewed with 75% of participants in the female group. However, our original sample and those who completed the screening instrument were also near 75% female. Figures 3 and 4 show more details regarding gender distribution of final participants and screener respondents.

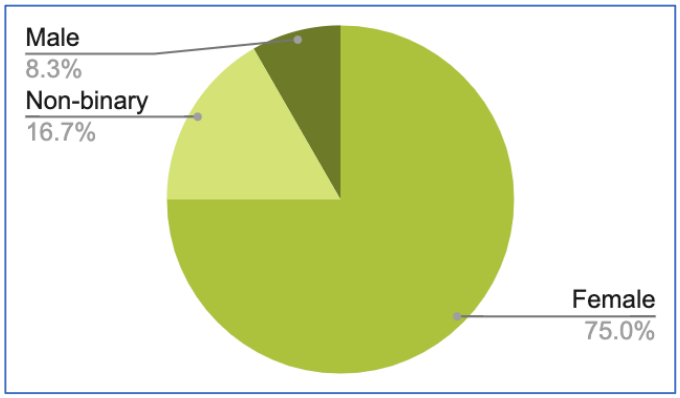


Figure 3. Gender Distribution of All Respondents (N=12)

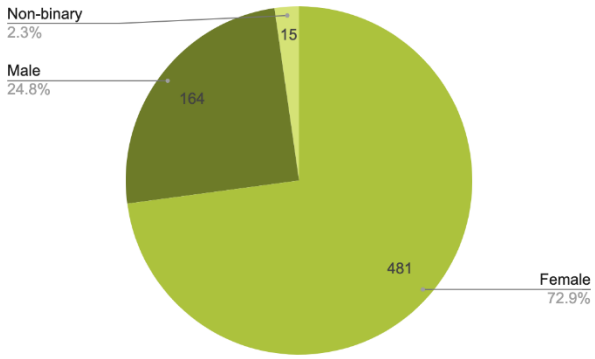


Figure 4. Gender Distribution of Eligible Screening Questionnaire Respondents (N=669)

Usability Session Instrument and Tasks

Twelve participants completed online usability sessions. The first two sessions were conducted with a convenience sample of pilot testers who met the eligibility criteria. Both participants resided in Western Washington; one was a new user and one an existing user. The goal of these two sessions, in addition to collecting usability data, was to refine the test instrument before using it with a wider participant group. Some minor adjustments were made to the script to improve session flow after pilot testing.

During the usability sessions, participants were first asked a series of questions about their technology use and their experience with vaccine verification (these questions varied based on User Status). Next, participants were asked to complete three short tasks on their smartphones, as listed in Table 2, below. The tasks were framed as scenarios during which users would interact with WA Verify: initial sign up and finding the records on a smartphone to use as verification documents for COVID-19 vaccine requirements. While completing each task, the moderator encouraged participants to "think out loud" and prompted them with specific questions about each stage of the process they were experiencing.

Table 2. Three WA Verify tasks used in usability sessions

Task 1	You are on your way to meet a friend at an event, they have just texted you that you will need to verify your vaccine record to enter the venue, and you don't have your paper card, or your typical method of vaccine verification. [NEW USER] You remember hearing about WA Verify [EXISITING USER] You remember having set-up WA Verify but have gotten booster shots and/or have gotten a new phone since then.
Task 2	Now that you have found the WA Verify site, please go through the process of [NEW USER] signing up with WA Verify [EXISTING USER] - updating your record with WA Verify
Task 3	You arrive at the event and need to provide proof of vaccination to enter and meet up with your friend. Use your phone to display the QR code or vaccine record.

Data Collection and Analysis

Analysis included data captured on the initial screener survey, recorded answers to questions asked at the beginning of the usability session, moderator notes and recordings, and transcripts of the usability sessions. Moderator notes were taken both during and after the usability sessions and focused on participants' experience of the WA Verify process (moderator notes are provided as a link in Appendix E). Transcripts were automatically generated during the Zoom sessions and were supplemented by the recordings. While the automated transcription software was not precise, transcripts were checked against and augmented by the audio and visual recordings. Quantitative data from the screening questionnaire along with session notes and quotes were synthesized, categorized, and aggregated into a spreadsheet. Because the sample was small, data were easily managed in the spreadsheet, which was also used as the primary analysis tool. Analysis of these data generated summaries of participant characteristics and performance reports describing WA Verify Tasks completed during the usability sessions.

Results

Participant Characteristics

Vaccine Verification Method

Participants reported using five different vaccine verification methods, the most common of which were the paper CDC COVID-19 Vaccination Card and WA Verify (Figure 5). Five participants (41.7%) used a single verification type, while 7 (58.3%) reported using two or more vaccine verification methods (Figure 6). Of those who using a single verification method, four used WA Verify and one had never been asked to verify their vaccination record before but indicated that, if asked, they would log into their online health portal to pull up vaccine records.

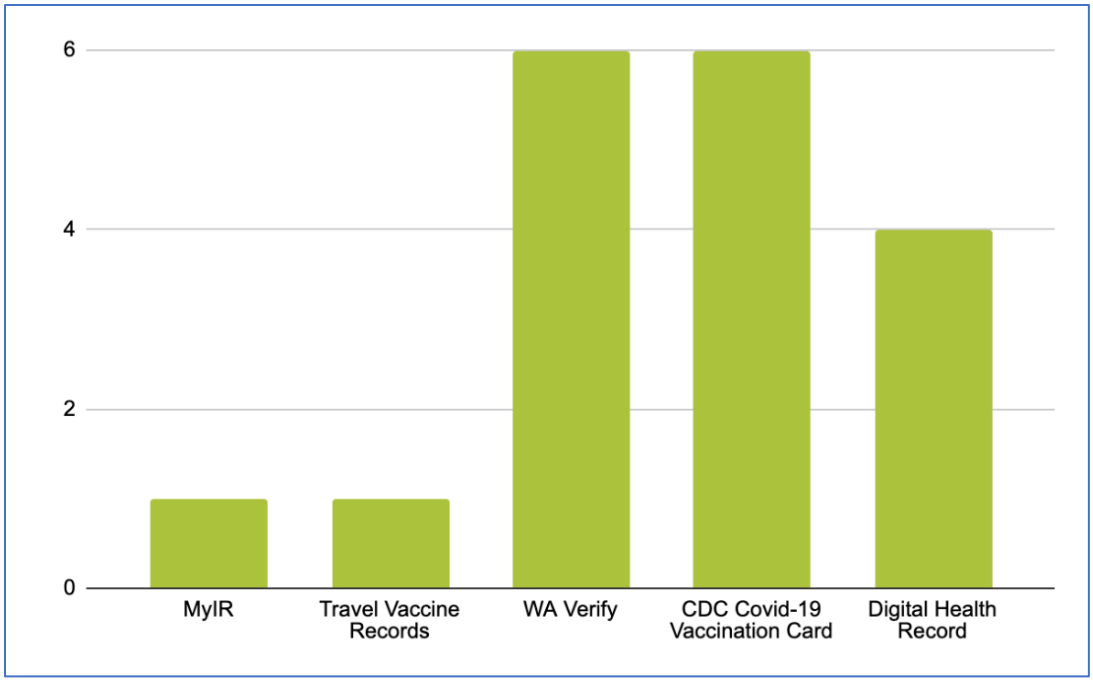


Figure 5. Histogram of Participants' Current Reported Vaccine Verification Methods

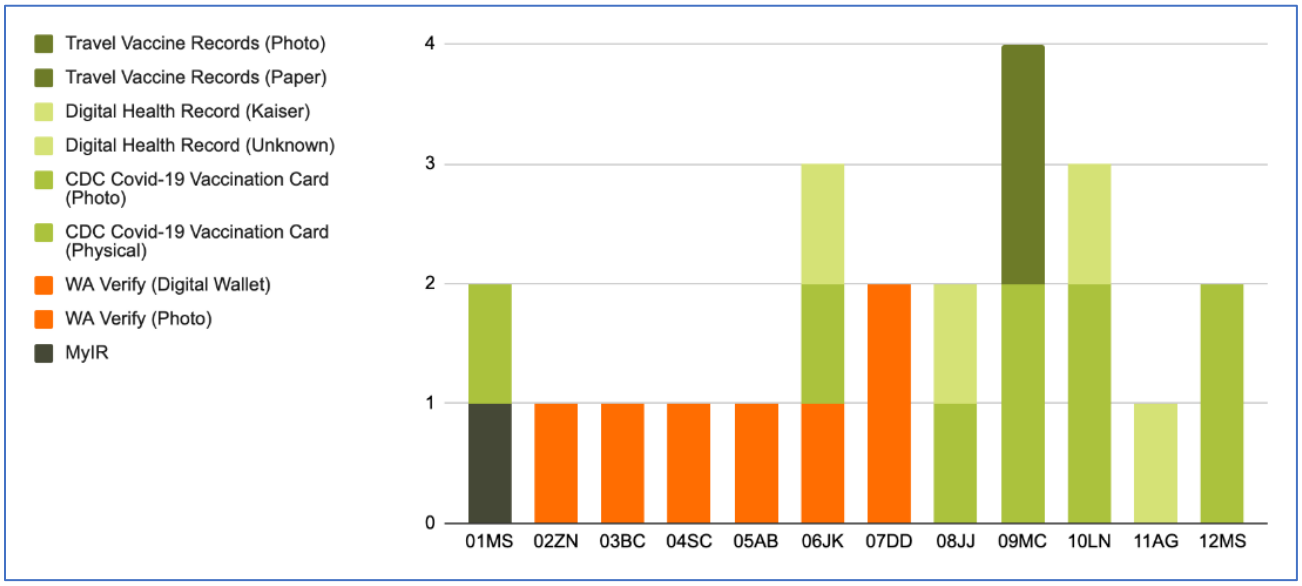


Figure 6. Current Verification Methods by Participant

Looking at verification methods by User Status, Figure 7 shows that existing WA Verify users had less variation in their current verification methods compared to those who had not adopted WA Verify.

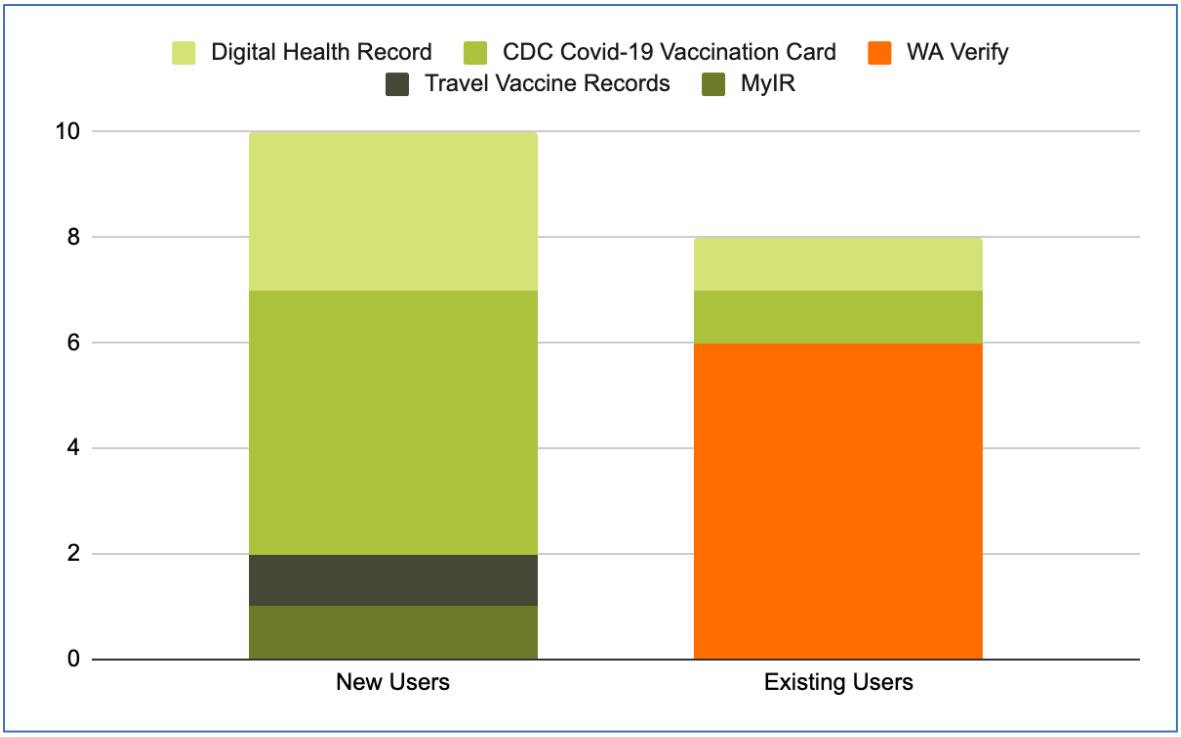


Figure 7. Current Verification Methods of New and Existing Users

Access to Another Person's Vaccine Record

Five of the six existing users were asked if another person's vaccine record was stored on their phone. Two of the five reported they had another person's vaccine record stored on their phone, and one user reported their vaccine record was stored on another person's phone. Users reported storing an additional person's record (close family members) on their phones, sometimes for convenience and sometimes because that person's age or disability made access to technology difficult. While none of the participants used WA Verify for a child's record, participant 12MS stated, "Why wouldn't parents use this with their kids and their vaccinations for school? It seems like we should be using it for everything."

WA Verify Usability Exercise Results

Task Completion

Eleven of the 12 participants successfully completed all usability tasks (Figure 8). One participant was unable to complete all 3 tasks (07DD) because the system could not find their record, likely due to multiple potential spellings of their name.

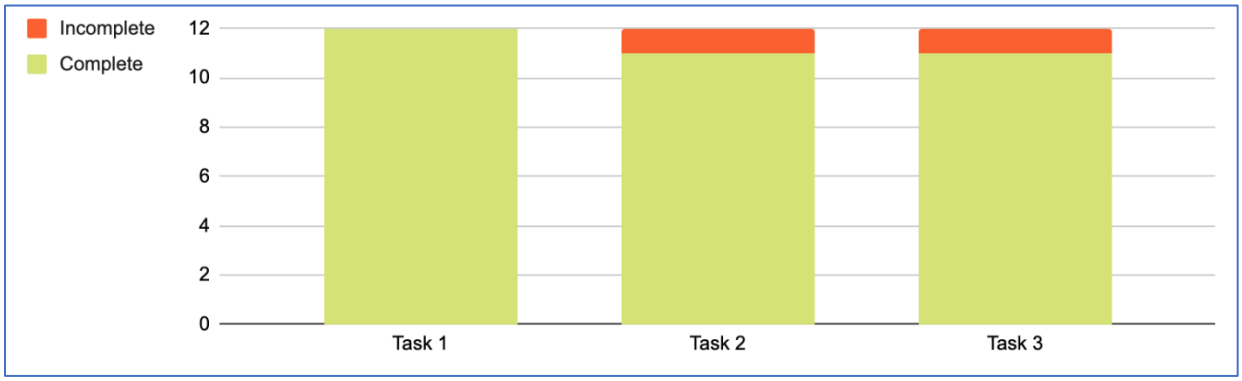


Figure 8. Count of Task Completion by Task

In Task 1, the only moderator assistance needed was a redirection of the participants from an app store search for WA Verify to a web browser search. In Task 2, the only participant who required significant moderator assistance was as mentioned (07DD) when trying to resolve the unmatched record. Other participants needed minor assistance, primarily to address issues with their devices, for example, assistance with the digital wallet feature which was not installed or up to date. The level of moderator assistance by task is illustrated in Figure 9.

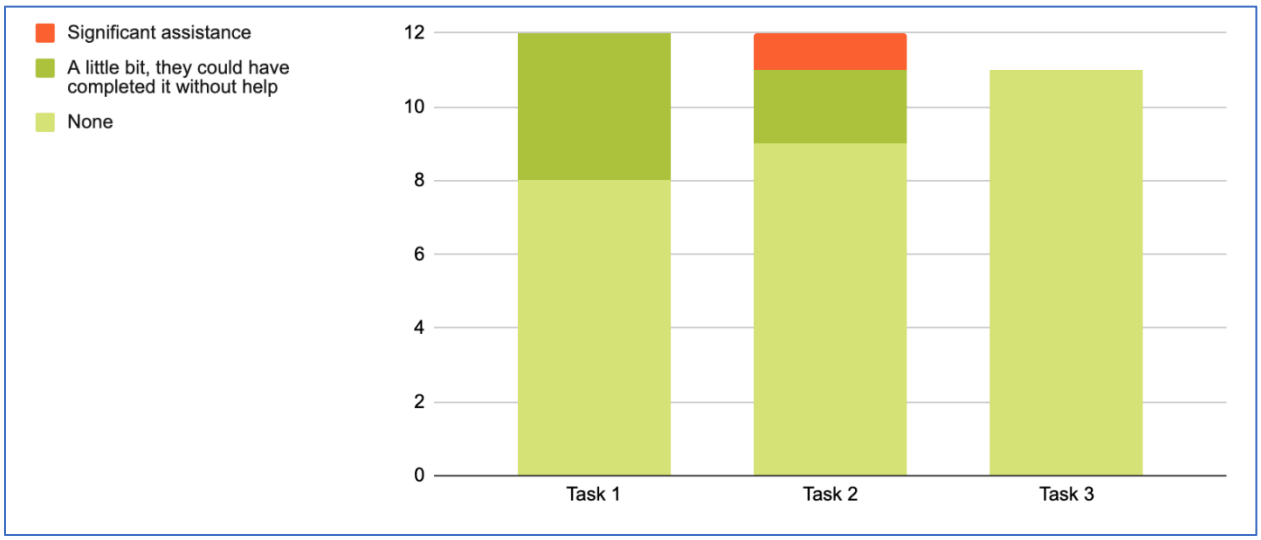


Figure 9. Moderator Assistance Required by Task

QR Code Storage Method

Seven out of the eleven participants who were able to complete the first two tasks chose to save the WA Verify QR Code to their phone's digital wallet. Three participants chose to download the QR code, and two out of the three of those participants who chose to download the QR code did not already use their phone's digital wallet. One participant chose to save the QR code to their phone's health app. Figure 10 shows the distribution of how participants chose to save the QR code.

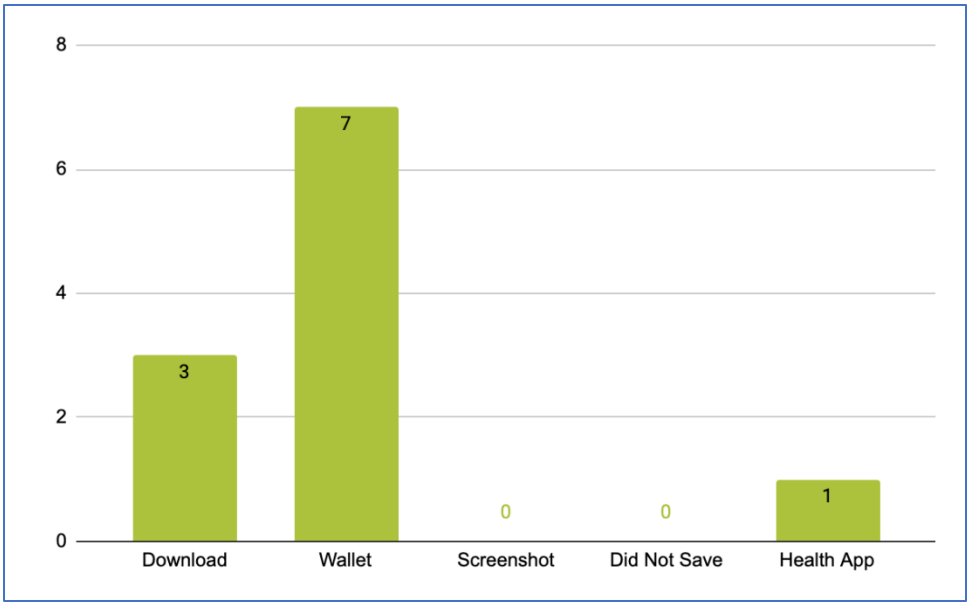


Figure 10. Methods of Saving WA Verify QR Code Among Those Who Completed the Task (N=11)

Future Use of WA Verify

Overall, participants' responses to using WA Verify were positive. All new users stated they intend to use the tool in the future for vaccine verification. Participant 04SC appreciated the speed of the set-up process, stating, "Oh, I think this is going to be great. This is really quick compared to what we had to do before." Regarding saving the record to a digital wallet, Participant 06JK stated, "I have already have a ton of stuff on my phone. So, if I can centralize things, I prefer to do that." Two participants expressed that WA Verify would be very useful for personal use in keeping track of their last booster shot date, in addition to using it as a verification tool (Table 3).

Table 3. Participant Descriptions of Personal Use of WA Verify

"I think it's really important, especially for those of us who have got a lot of vaccines, and we start to lose track of it can also act as a reminder to me. Sometimes I forget, I'll read that another vaccine is available, a booster, and this has to be like 3 months since your last one, and I'll be thinking when was my last one, and then usually I have to go to my provider, and kind of go through my records to find it. But this would be a lot simpler to know when was the last one I had. So, I think it'd be a useful tool for me not just to show somebody else."	"It's very aesthetically organized, which I think is helpful people can immediately see what dose when? And as someone who works in healthcare, too, I might even use it for my own reference, and it's nice to just say for the manufacturers and what the dose was in one spot."
Participant 11AG	**Participant 09MC**

Recommendations

Analysis of the usability exercise data revealed seven medium-to low-priority recommendations to

improve the usability of WA Verify. These recommendations are summarized along with associated findings in Table 4., below. Each recommendation, associated findings and other details are explored in in this section.

Table 4. Summary Listing of Recommendations and Associated Findings

Recommendation	Associated Finding(s)
1. Clarify communications and FAQs on the website regarding the WA Verify platform, how to enroll and how it works.	*Participants thought WA Verify was an App and looked for it in the App store or Play store.*
2. Reduce density of the text on the WA Verify website or reformat the content using bullets or visual displays to increase readability.	*Most participants either skimmed or skipped the text on the WA Verify Website.*
3. Include the phrase "Washington State Immunization Information System" in the website's introductory paragraph and provide more information on how immunization information is saved, accessed, and stored on the FAQ page.	*Participants wanted more information on how WA Verify was accessing their vaccine records.*
4. Improve user expectations by adding information regarding the average time to receive a text or email in both the introduction and confirmation sections of the website.	*Participants assumed the 24-hour period referenced was the average time to receive the text/email.*
5. Ensure authority of the SMS message by highlighting the source, i.e., the message is sent from WA Verify and the Washington State Department of Health.	*Participants determined the message's validity through context, but their phones highlighted it as suspicious.*
6. Inform users that a screenshot is an additional method of saving the QR code.	*Participants were unaware that screenshotting the QR code was a valid method of saving when they did not want to use the other offered methods.*
7. Reformat the downloadable version of the WA Verify QR code to include room for multiple boosters and continue testing to determine exactly how many users this affects.	*One participant did not have a valid QR code upon downloading as their number of boosters exceeded the screen size of their phone.*

1. Clarify the format of WA Verify (App vs. Web Page)

The most common point of confusion for participants was where to find and begin the WA Verify set-up process. Five out of all twelve participants went first to their phone's app store, thinking the system would be app-based rather than begin at a DOH website. Figure 11 shows participants' first attempt to locate and begin the WA Verify set-up process. Participant 09MC stated, "so there's no app, that was confusing". All participants who started in the app store next went to their web browser to find WA Verify with little to no prompting. While this did not lead to additional issues with usability participants, it did cause confusion and might discourage some potential users.

Recommendation: Clarify communications and FAQs on the website regarding the WA Verify platform, how to enroll and how it works.

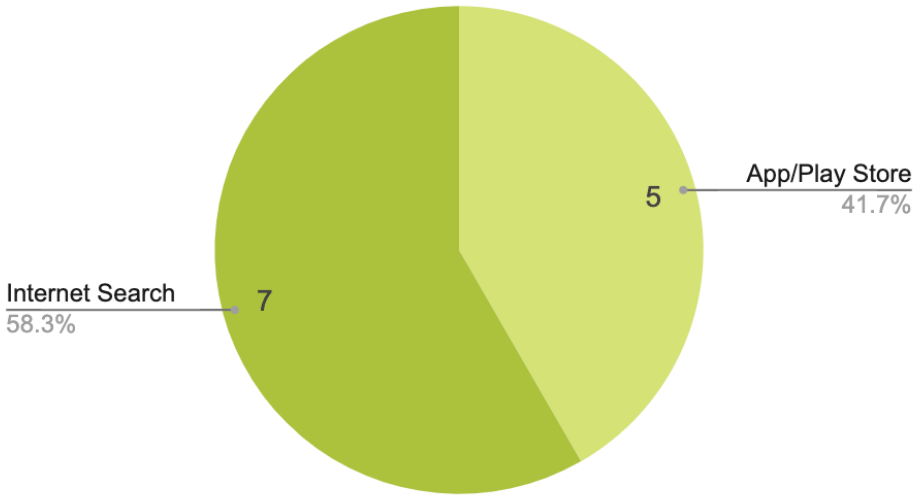


Figure 11. Participants' initial method of finding WA Verify (N=12)

2. Improve readability in introductory paragraphs on the WA Verify website

Only one of the participants read the introductory paragraphs on the website completely without any prompting. As seen in Figure 11, seven participants skimmed the text, while four skipped it entirely. Those participants who read or skimmed the content all found useful information within the text.

Recommendation: Improve readability by reducing the text density or reformatting the content as bullet points or visual displays.

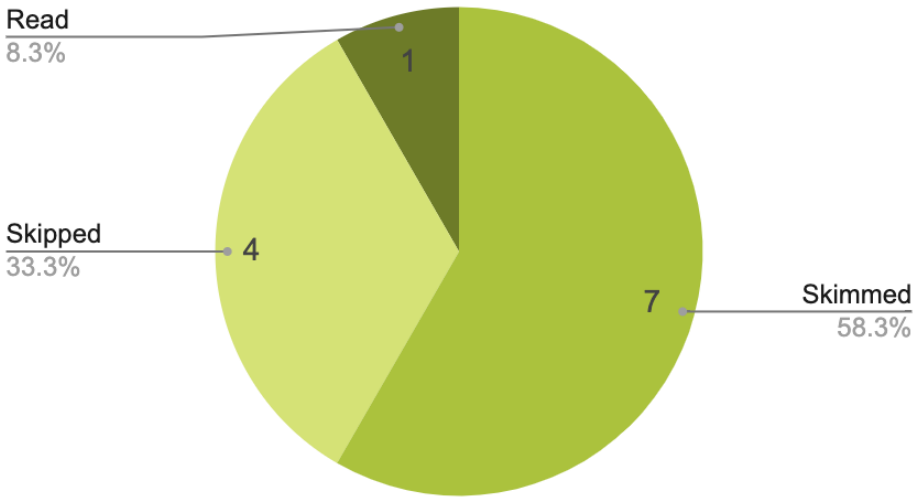


Figure 11. Participant Reading Depth of Introduction Content (N=12)

3. Provide information about where the immunization data comes from

Two of the participants wanted more details on how their contact information and medical records were linked and where the immunization record data came from. The FAQ page has a brief description:

> "The Digital COVID-19 Verification Record system draws COVID-19 records from the state's immunization systems. Enter your name, date of birth, and an email or mobile phone number

associated with your vaccination record, then create a four-digit PIN number. If the information you submitted matches the official record, you will receive a text or email with a link to your Digital COVID-19 Verification Record. Enter the PIN you created to view the record."
source: waverify.doh.wa.gov/faq

However, the FAQ primarily provides information already covered on the main page. It would be helpful to add more detail about the "state's immunization systems" to the text on the main page. In addition, more information about how the vaccination data is saved, accessed, and stored could be included on the FAQ page. Alternatively, part of the descriptive text on the main page could be replaced or augmented with a graphic (for example, Figure 12).

Recommendation: Include the phrase "Washington State Immunization Information System" in the website's introductory paragraph and provide more information on how immunization information is saved, accessed, and stored on the FAQ page.



Figure 12: Simplified Diagram of the WA Verify Signup Process

4. Provide information about average wait times to receive message with link (email and text)

Multiple participants were confused or responded negatively to the 24-hour window mentioned on both the introduction and confirmation pages ("you can expect to receive your link automatically within the next 24 hours", highlighted in *Figure 13*). When they were presented with the scenario of being on the way to the event there was concern that they would not be able to finish the process before they "arrived". Participant 09MC specifically mentioned that they wished the information presented to them had said how long it typically takes to get the text, not just when to worry about having not received it. Additionally, both participant 11AG and participant 04SC indicated surprise with how fast the text arrived, with 11AG stating that "That 24 hours is more like 24 seconds".

Recommendation: Improve user expectations by adding information regarding the average time to

receive a text or email in both the introduction and confirmation sections of the website.

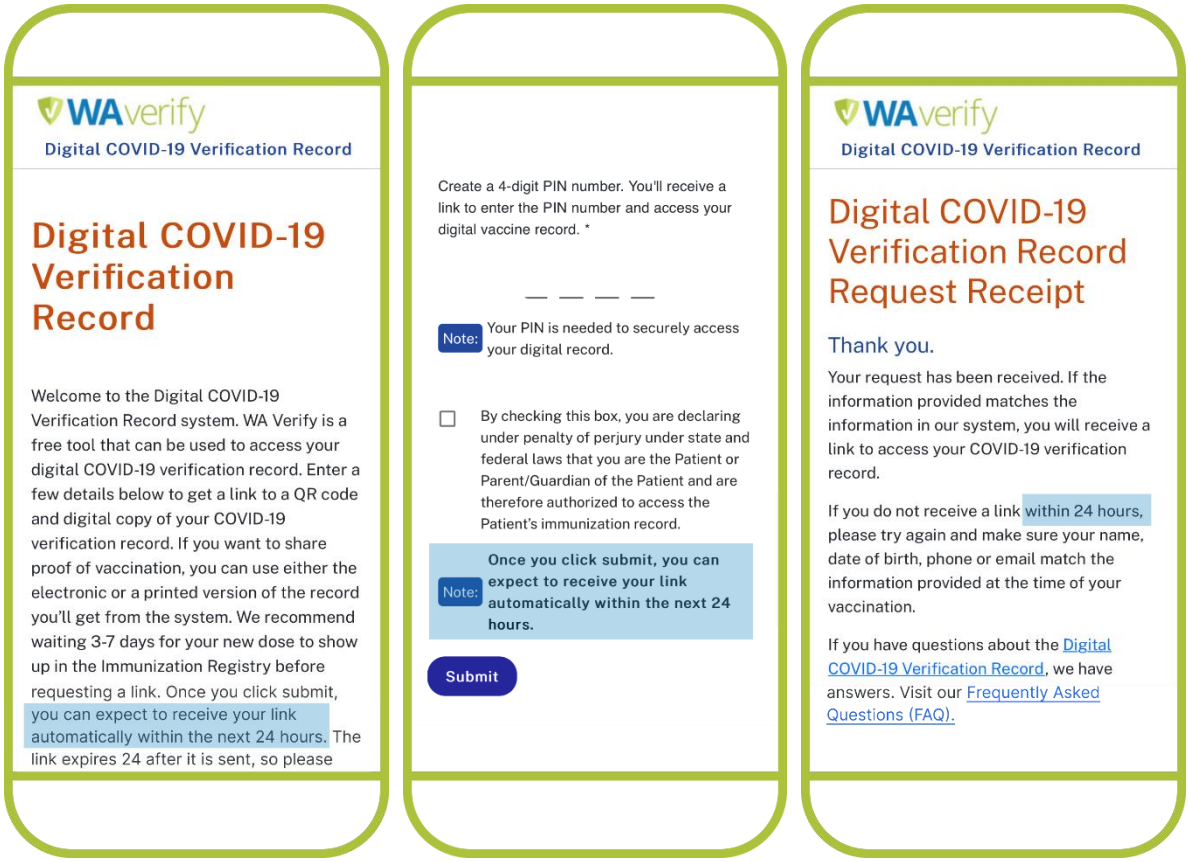


Figure 13. 24-Hour Verbiage Highlighted in Blue

5. Provide more information/options for saving the QR code

As noted previously in *Figure 10*, none of the participants chose to screenshot the displayed QR code, but some noted that they would have appreciated that option being included in the "To Save" options. Participant 10LN had had problems saving the QR code to their phone using the two displayed options, shown in *Figure 14*. They had not set up their phone's digital wallet or health app and their phone only offered them, as they said "clunky", options such as saving to Google Drive. It is recommended that text be added to the save page to inform users that a screenshot of the QR code is an also valid method for saving the information for later use.

Recommendation: Inform users that a screenshot is an additional method of saving the QR code.

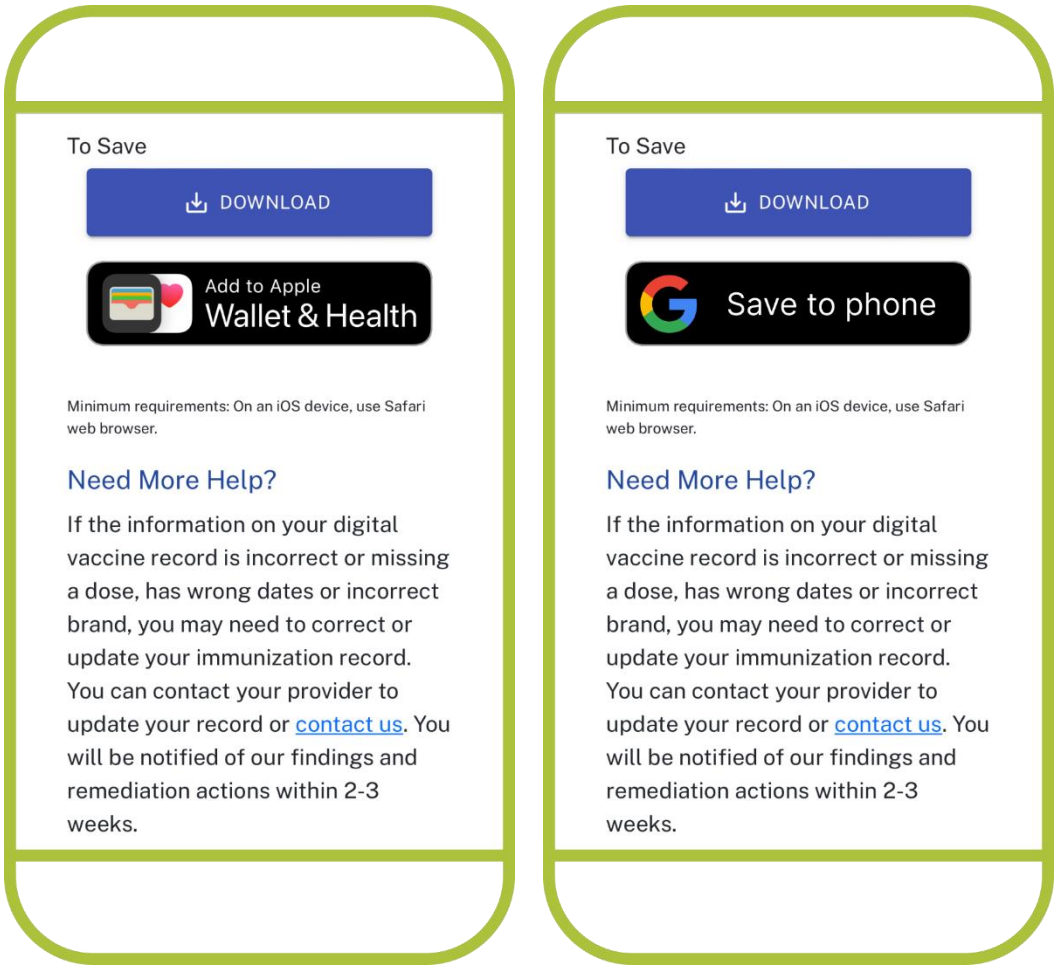


Figure 14. Methods of Saving QR Codes

6. Clearly identify incoming text messages as coming from DOH/WA Verify

Overall, and primarily due to the immediate timing of the message, the participants did trust that the message they received was from the WA Notify system. However, there were a few elements they noted that gave them pause. First, unlike the email which included the WA Verify logo, the text message only had the WA Verify name within the URL sent in a separate message, with the message only referring to the "Digital COVID-19 Verification Record system". Some participants recognized this name from the site, but as covered in the section above, only 8.3% of participants read all the text on the site. Participant 10LN specifically noted that the wording of the text message does not mention WA Verify or Washington State DOH anywhere and they had to look in the URL to verify the source. Second, the text came from a number starting with 1 (8...) which some participants associated with marketing communications. Third, phone systems call out the messages as Junk or Spam messages. These issues are all shown in *Figure 15* and *Figure 16* below. While not all these issues might be able to be addressed, it is recommended to change the text of the text message to highlight that the message is coming from WA Verify and Washington DOH.

Recommendation: Ensure authority of the SMS message by highlighting the source, i.e., the message is sent from WA Verify and the Washington State Department of Health.

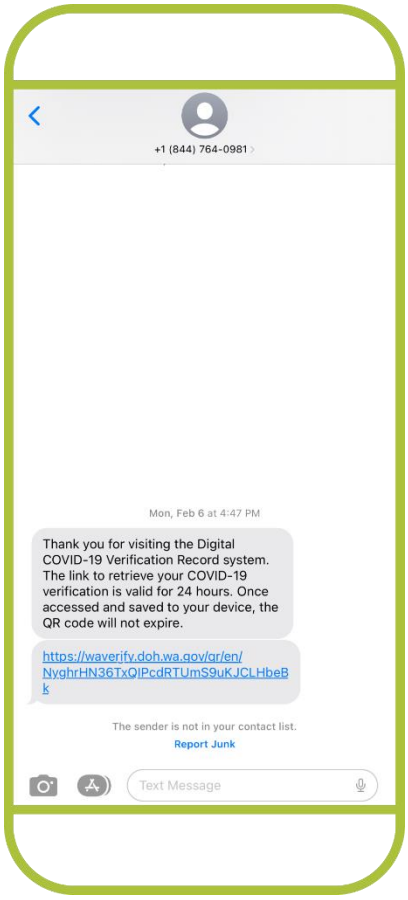


Figure 15. WA Notify Text Message on iPhone

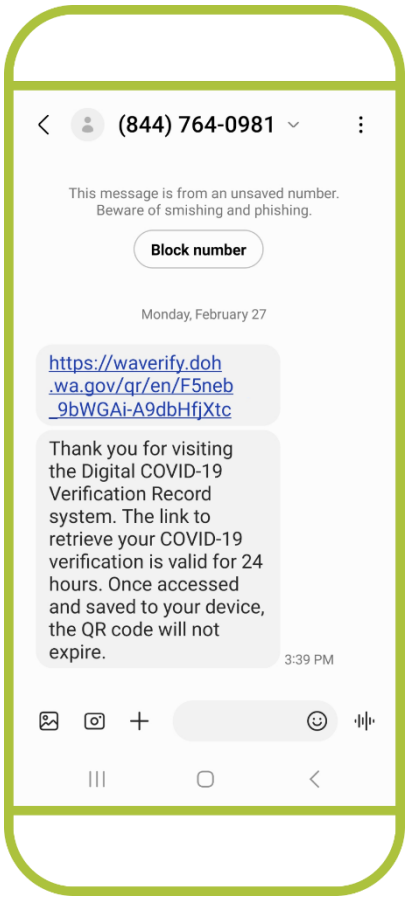


Figure 16. WA Notify Text on Android

7. Address problems displaying large numbers of vaccines

Users with over five vaccines downloading the QR code to Images may not be able to access a useable QR code. In this exercise this issue only affected Participant 11AG, but it would be a problem for any WA Verify user with their characteristics. Participant 11AG is 60+ and a front-line worker, so has been eligible for every round of COVID-19 vaccines and boosters; on the day of the interview, they had five doses listed on their vaccine record. The image file that was downloaded was longer than the phone was capable of displaying, cutting off the top 1/3 of the QR code and their 5th booster shot. This issue is replicated as a mockup in *Figure 17*.

Through further testing, it was determined that this issue won't affect all users with high numbers of booster shots, as it highly depends on how their phone's operating system chooses to display images that are longer than the phone's screen for those users who it does fit to the width rather than the height, it significantly reduces the usability of the WA Verify system. It is recommended that the display of the dose dates be reformatted to fit larger quantities of booster shots, both to solve this issue and to preemptively account for additional records as people continue to get their annual booster shots.

Recommendation: Reformat the downloadable version of the WA Verify QR code to include room for multiple boosters and continue testing to determine exactly how many users this affects.

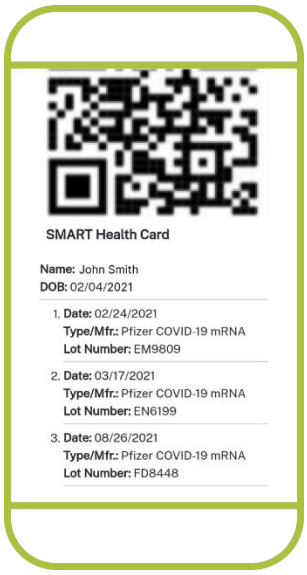


Figure 17. Mock-up of the cut off QR Code

Conclusion

Digital public health tools like WA Verify have the potential to enhance the conduct of public health surveillance and response. However, challenges in their deployment, acceptance and adoption, communication of their value, and assessment of the equity issues involved in adoption of these technologies among disproportionately affected groups and vulnerable populations need to be identified and addressed. Uncovering these challenges and generating approaches to addressing these issues can be informed by conducting usability exercises such as reported here. While no major usability concerns were identified and the recommendations generated from this exercise consist primarily of text content and formatting changes within the WA Verify website and the text message communication with users, findings from this exercise form a foundation for future usability exercises with a larger sample. Continued and expanded usability evaluations can provide valuable input into the design and development of digital public health tools and ensure their sustained adoption, acceptance and use for the benefit of all Washington State residents. As DOH explores new and expanded use cases for WA Verify and other digital public health tools, it will be important to ensure these technologies meet user expectations and usability priorities.

APPENDIX

Appendix A. WA Verify UX Questionnaire Recruitment Email

<SENT 2/26/23 5:30PM>

Greetings,

You are receiving this email because in November 2021, you completed a survey about using WA Notify, Washington State's smartphone-based COVID-19 exposure notification tool. Your input from that survey helped our evaluation team here at the University of Washington develop recommendations to improve digital public health tools and ensure these new tools will help protect Washington residents and their communities.

At the end of that survey, we asked if you would be willing to participate in future activities. We are currently evaluating another digital public health tool, WA Verify—Washington State's digital vaccination verification tool—and are hoping you could again provide input. The activity this time is an online usability study that will take approximately 30 minutes of your time. If you are selected to participate in this activity, you will be reimbursed with a $50 Amazon gift card.
To let us know of your interest, please complete a short questionnaire at: [survey-link]. If this link does not work, please try copying the link below into your web browser: [survey-url]
The questionnaire should take approximately 2 minutes to complete. We will email you before March 9th to let you know if you've been selected to participate in the usability session. Your participation in the usability session is completely voluntary, and your responses will be kept anonymous.
If you have any questions about the questionnaire or the usability session, please contact our team at waverify@uw.edu.

In advance, thank you very much for your interest and participation. Your input will help improve digital public health tools in Washington.

Sincerely,

The University of Washington Digital Public Health Tools Evaluation Team

Appendix B. Screening Questionnaire

1. Do you currently live in Washington State?
    a. Yes
    b. No [Ineligible]
2. Which county do you live in?
    a. [Drop down selection listing all WA Counties]
3. Do you have a smartphone?
    a. Yes - Apple
    a. Yes - Android
    b. Yes – Other
    c. No [Ineligible]
2. Have you received any COVID-19 vaccines in Washington State?
    a. Yes
    b. No [Ineligible]
3. To participate in the usability session you will be doing an activity on your smartphone **while** you are connected with us through Zoom on a separate device.
   Do you have a device in addition to your smartphone that you can use to connect to the Internet for our Zoom session?
    a. Yes
    b. No [Ineligible]

FIELD NOTE: *In addition to your smartphone,* a *tablet, laptop or desktop computer that connects to the internet will work for the usability session.*

4. How old are you?
    a. 18 – 29
    b. 30-39
    c. 40-49
    d. 50-59
    e. 60-69
    f. 70-70
    g. 80+
5. What is your highest level of education?
    a. Less than high school
    b. GED (General Educational Development)
    c. High school graduate
    d. 2-year degree or some college
    e. 4-year degree or more

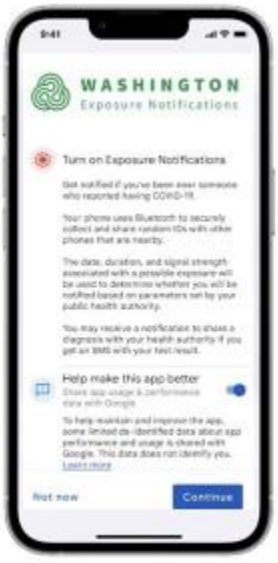
6. Are you…?
    a. Female
    b. Male
    c. Transgender
    d. Non-binary/non-conforming

e. Prefer not to respond

The Washington Department of Health has two COVID-19 systems for smartphones with similar names.

WA Notify is a COVID-19 Bluetooth exposure notification tool and will notify you if you have been in close proximity to a person who also uses WA Notify and tested positive for COVID-19.

WA Verify is a tool for keeping **your COVID-19 vaccination information** on your phone. WA Verify provides a QR code and text record of your COVID-19 vaccinations so that you can share your vaccination information when needed.



WA Notify



WA Verify

7. Do you have WA Verify on your phone?
   a. Yes
   b. No
   c. Not sure

   FIELD NOTE: *WA Verify is the Washington State Vaccine Verification Tool*

Appendix C. Test Script

Pre-conditions:

- ☐ User has confirmed in writing their willingness to be recorded.
- ☐ User has mobile phone on and connected to the internet.
- ☐ Interviewer is logged in to Zoom, video and audio are tested and on.
- ☐ User is logged in to Zoom with a PC, laptop or tablet, audio and video are tested and on
- ☐ User is logged in to Zoom with their phone (muted, camera off, sharing screen)
- ☐ Interviewer has read consent text and confirmed that user consents to the session being recorded.
- ☐ Zoom recording has started.

Hello! My name is _____ and I am a user researcher at the University of Washington. Thank you so much for joining me for this User Experience session. First, do you consent for the usability session to be **recorded** and for your anonymous data to be used to improve public health systems and in reports or articles?

The purpose of our session today is to review the tool called WA Verify and to get your opinions about how it works and how you feel about it. This session will be focused on three short tasks on your phone related to vaccines and providing proof of your COVID-19 vaccines. I am going to be here to guide you if you need help, but I would like to first see if you are able to figure out the way to work with the tool to complete the tasks. I want to emphasize that we are testing the application, and not you. I did not design the application and I want your honest feedback to improve it. While you are working, you can "think aloud" and say if you are confused or find anything distracting, confusing, missing, excessive, helpful, etc. All of those thoughts spoken aloud would be very helpful for me to know. If a question pops to mind, say it! There are no comments that are off limits and no stupid questions!

Do you have any questions about what you should do or what will happen during the session?

OK. Let's get started!

First, I'm going to ask you a few quick questions to get an idea of your typical technology use and your experience with vaccine verification.

[1] Pre-Task Questions:

Overall, how confident do you feel using computers, smartphones, or other electronic devices to do the things you need to do online?

Do you use the digital wallet on your phone (Apple Wallet/Google Wallet)?

1.1. (If Yes) What information do you store in your digital wallet?

1.2. (If No) Why not?

2. When you have had to verify (provide proof of) your COVID-19 vaccination status, what have you used?

[2] New User Pre-Task Questions:

1. At which type of venues or destinations have you needed to verify your vaccination status? (Travel, Dining, Entertainment, Other?)
    1.1. Since the beginning of the pandemic?
    1.2. Recently, in the past 6 months or so?
2. WA Verify is a tool provided by the Washington State Department of Health to allow you to carry and show your vaccine record on your phone instead of carrying the paper vaccine card. Before receiving this survey invitation, had you heard about WA Verify?
3. Did you immediately know what WA Verify was or did it take a minute to figure out which of the Washington State public health tools we were referring to?

[3] Existing User Pre-Task Questions:

1. When did you first set up WA Verify?
2. Where/How did you save the WA Verify QR code when you first set it up? Can you show me?
3. How frequently did you use WA Verify when you first set it up? *(Daily, Weekly, Monthly, Irregularly, Never?)*
4. How frequently do you currently use WA Verify? (*Daily, Weekly, Monthly, Irregularly, Never?)*
5. Do you use WA Verify for more than just yourself on your own device?

(If Yes) Who else have you had on your device?
At which type of venues or destinations have you used WA Verify? (Travel, Dining, Entertainment, Other?)

[T1] Task #1:

Here is the scenario of the first task I will ask you to walk through:

You are on your way to meet a friend at an event, they have just texted you that you will need to verify your vaccine record to enter the venue, and you don't have your paper card. You [NEW - remember hearing about WA Verify] [EXISITING – remember having set up WA Verify but have gotten booster shots since then / you've gotten a new phone].

1. How would you go about figuring out how to set up WA Verify on your phone?

Moderator Questions for Task #1

1. Did the user complete the task? (Did the user find the WA Verify site?)
    a. Yes
    b. No
2. How much assistance did the user need to complete the task?
    a. None
    b. A little bit, they could have completed it without help
    c. Significant assistance
    d. They needed guidance the whole way
3. Moderator Notes: make notes of areas of confusion, questions spoken aloud and comments and the page of the system they were on at the time.

[T2] Task #2: Get your vaccine record on your phone as a QR code.

Now that you have found the WA Verify site, please go through the process of [NEW - signing up] [EXISTING - updating your record] as far as you can. Again, if you have comments along the way, please say them out loud! I can help you if you need it.

<Send link if needed: https://waverify.doh.wa.gov/>

<user will move through the process; interviewer will be available to help with a slight delay (wait 5(?) seconds before responding to a question>

Structured interview questions for Task 2. The moderator will ask the questions aloud and record on paper or through redcap either during or after the interview. See Appendix C for page definitions.

1. **Page 1 – Intro**
    1.1. What do you think about the contextual information provided at the top of this page?
    1.2. People have different comfort levels with sharing personal data online. How comfortable are you with sharing the personal information requested on this page?
    1.3. Did you choose to receive a text or email message?
    1.4. Do you have any other feedback about this step of the process?
2. **Page 2 – Confirmation**
    2.1. Do you have any other feedback about this step of the process?
3. **Email/Text Message**
    3.1. How clear is it to you that the message you received was from the WA Verify system?
    3.2. How trustworthy is the message you received?
    3.3. Do you have any other feedback about this step of the process?
4. **Page 3 – Pin**
    4.1. Were you able to successfully enter your PIN and see your QR code?
    4.2. Do you have any other feedback about this step of the process?
5. **Page 4 – Save**
    5.1. How did you save the QR code to your phone?
        5.1.1. Why did you choose that method?
    5.2. Do you have any other feedback about this step of the process?

Moderator Questions for Task #2

1. Did the user complete the task? (Does the user have their vaccination record on their phone in the form of a QR code from WA Verify? Can be saved as a downloaded image or in the digital wallet.)
    1.1. Yes
    1.2. No
2. How much assistance did the user need to complete the task?
    2.1. None
    2.2. A little bit, they could have completed it without help
    2.3. Significant assistance
    2.4. They needed guidance the whole way
3. Moderator Notes: make notes of areas of confusion, questions spoken aloud and comments and

the page of the system they were on at the time.

[T3] Task #3: Pull up your vaccine record/QR code.

<start fresh/go to home screen>
Continuing the scenario, you arrive at the event and need to provide proof of vaccination to enter and meet up with your friend. Use your phone to display the QR code or vaccine record. If you have questions or comments, please say them out loud. I can help you if you get lost or confused.

Structured interview questions about Task# 3. The moderator will ask the questions aloud and record on paper or through redcap either during or after the interview. See Appendix C for page definitions.

1. Vaccine Verification
    1.1. Were you able to locate and display your vaccine QR code?
    1.2. How confident would you be finding and displaying the vaccine QR code in a real world scenario?
    1.3. Do you have any other feedback about this step of the process?

Moderator Questions for Task #3

1. Did the user complete the task? (Did the user successfully find their digital vaccine record and display it? Was it scannable using the verifier?)
    1.1. Yes
    1.2. No
2. How much assistance did the user need to complete the task?
    2.1. None
    2.2. A little bit, they could have completed it without help
    2.3. Significant assistance
    2.4. They needed guidance the whole way
3. Moderator Notes: make notes of areas of confusion, questions spoken aloud and comments and the page of the system they were on at the time.

[4] New User Post-Task Questions.

1. Have you ever considered using WA Verify before today?
    1.1. (If Yes) Why had you not already set it up?
    1.2. (If No) Why did you choose not to use WA Verify?
2. Now that WA Verify is on your phone, how likely is it that you would use it if/when you are asked for proof of vaccination?
    2.1. (If Yes) How frequently do you see yourself using WA Verify now that it is on your device? (Daily, Weekly, Monthly, Irregularly, Never?)
    2.2. (If No) What are your hesitations about using WA Verify?

[5] Existing User Post-Task Questions:

1. Have you saved a new QR code since your most recent booster shot (not counting today)?
2. Before today, did you know that you needed to update your QR code to reflect your booster status?

Wrap Up:

This concludes our activity. Do you have any other feedback to share with us about the system?

We will be emailing you a digital amazon gift card within the next few days. Thank you for your participation, we appreciate it!

Appendix D. Participants

Spreadsheet available at Participants.xlsx

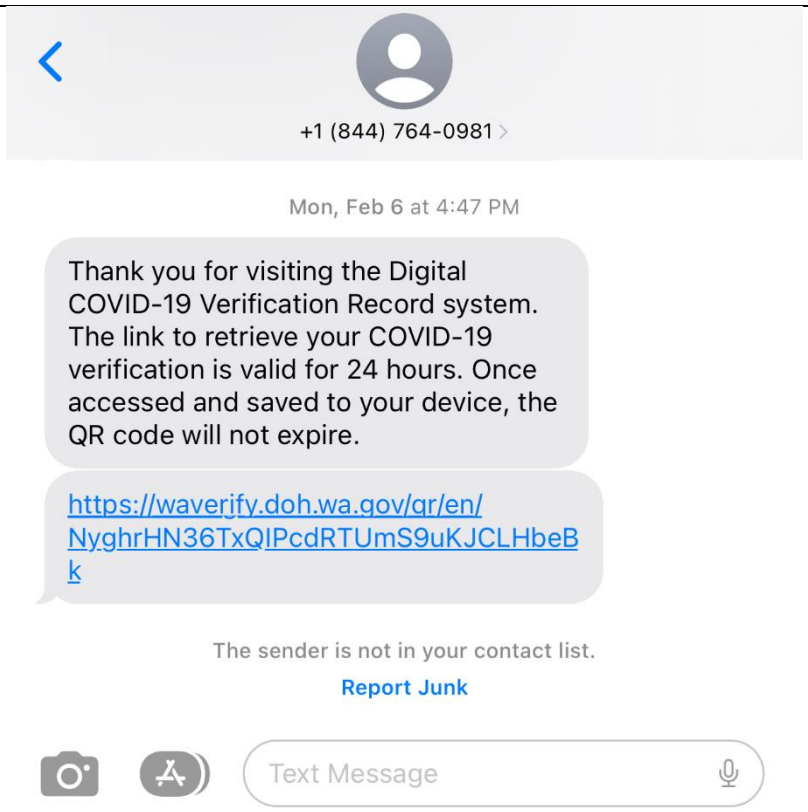
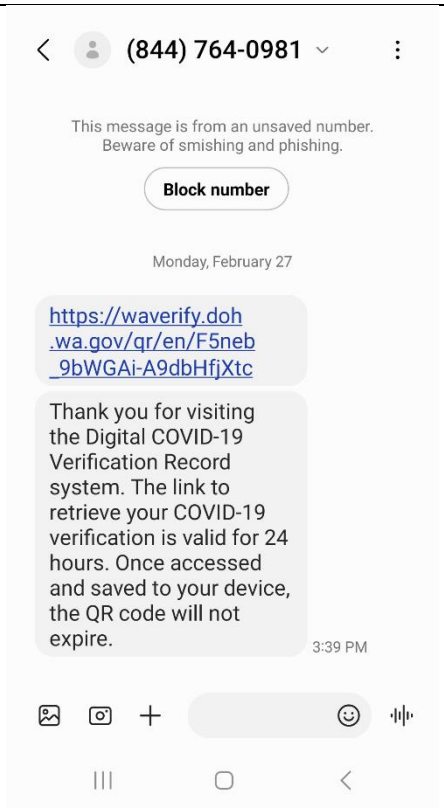
Record ID	Do you currently live in Washington State?	Which county do you live in?	Do you have a smartphone?	Have you received any COVID-19 vaccines in Washington State?	Zoom session?	How old are you?	What is your highest level of education?	Are you...?	Do you have WA Verify on your phone?
2000	Yes (1)	King (West)	Yes - Android (2)	Yes (1)	Yes (1)	18-29 (1)	4-year degree or more (5)	Non-binary/non-conforming (4)	Yes (1)
65	Yes (1)	King (West)	Yes - Android (2)	Yes (1)	Yes (1)	30-39 (2)	4-year degree or more (5)	Female (1)	No (2)
76	Yes (1)	Whitman (East)	Yes - Android (2)	Yes (1)	Yes (1)	60-69 (5)	4-year degree or more (5)	Female (1)	No (2)
1425	Yes (1)	Snohomish (West)	Yes - Apple (1)	Yes (1)	Yes (1)	30-39 (2)	4-year degree or more (5)	Male (2)	Yes (1)
563	Yes (1)	Thurston (West)	Yes - Apple (1)	Yes (1)	Yes (1)	18-29 (1)	2-year degree or some college (4)	Female (1)	Yes (1)
1018	Yes (1)	Pierce (West)	Yes - Android (2)	Yes (1)	Yes (1)	40-49 (3)	2-year degree or some college (4)	Non-binary/non-conforming (4)	Yes (1)
912	Yes	Spokane (East)	Yes - Apple	Yes	Yes	18-29	4-year degree or more	Female	Yes/No
1053	Yes	Yakima (East)	Yes - Apple	Yes	Yes	30-39	4-year degree or more	Female	No
83	Yes	King (West)	Yes - Android	Yes	Yes	18-29	4-year degree or more	Female	No
1234	Yes	Grant (East)	Yes - Apple	Yes	Yes	60-69	2-year degree or some college	Female	No/Yes
841	Yes	Spokane (East)	Yes - Android	Yes	Yes	60-69	4-year degree or more	Female	No
369	Yes	Yakima (East)	Yes - Apple	Yes	Yes	50-59	4-year degree or more	Female	Yes

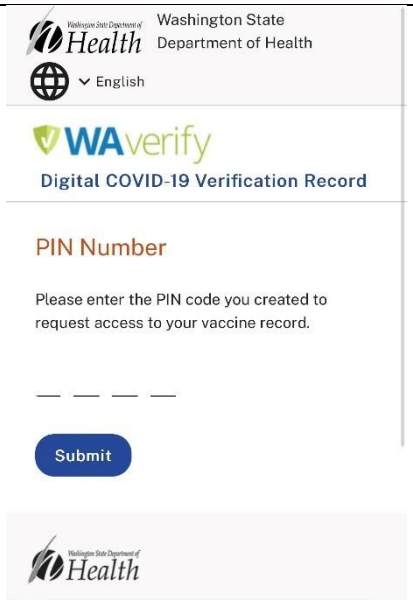
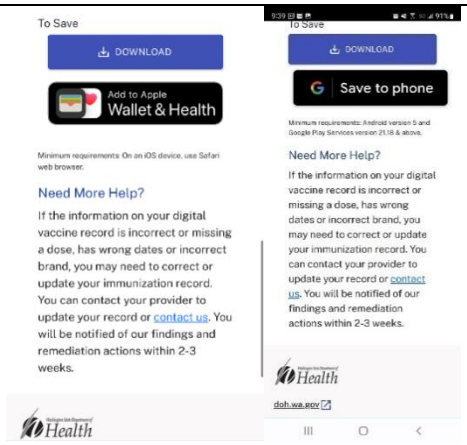
Appendix E. Session Notes

View session notes at Notetaking.xlsx

Appendix F. Page Definitions

Page 1 - Intro	
Page 2 - Confirm	

Text Message - iPhone	
Text Message - Android	

Email	WAverify **Digital COVID-19 Verification Record** Thank you for visiting the Digital COVID-19 Verification Record system. The link to retrieve your COVID-19 verification record code is valid for 24 hours. Once accessed and saved to your device, the QR code will not expire. View Verification Record Learn more about how to protect yourself and others from the Centers for Disease Control and Prevention. **Have questions?** Visit our Frequently Asked Questions (FAQ) page to learn more about your Digital COVID-19 Verification Record. **Stay Informed.** View the latest information on COVID-19.
Page 3 - Pin	
Page 4 - Save	

Vaccine Verification	 